Economic freedom and firm entry–exit decisions

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https://github.com/gsaco/ai-project

1 Question and track

Why might an institutional deterioration cause a larger output loss than the gain from an equal improvement? My thesis studies real GDP per capita after economic-freedom changes (Figure 1). In this Track B project, I model a firm that may close after a temporary setback. Even after institutions recover, rebuilding may cost more than future earnings. The aim is to explain a possible direction of asymmetry, rather than estimate the size of the GDP responses.

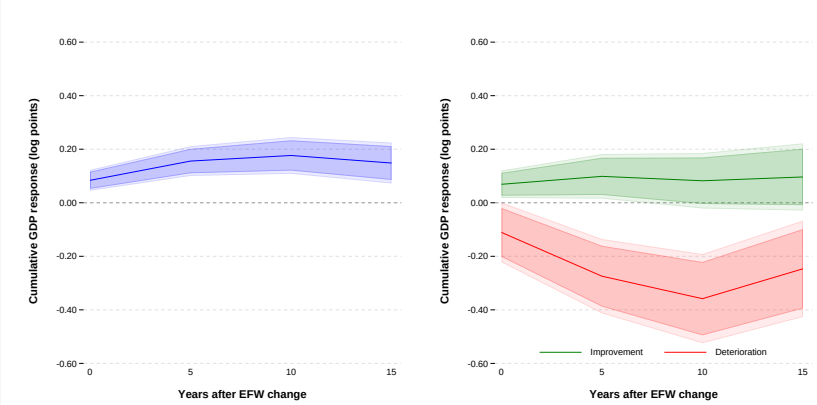


Figure 1. Thesis evidence. Cumulative log real GDP per capita responses to one-point EFW changes, combining current and once-lagged effects. Left: symmetry imposed; right: asymmetry allowed. Inner bands: 95% country-clustered pointwise intervals; outer: 95% simultaneous sup-t bands.

2 Baseline, changed assumption, and firm problem

Baseline: restarting is free, $\kappa = 0$. **Extension:** restarting requires a sunk cost, $\kappa > 0$. Technology, prices, other costs, and institutional paths are identical, isolating the effect of restarting costs. The mechanism follows Dixit (1989); here the firm knows when institutions recover.

Economic freedom concerns property use, contracting, and exchange. The Economic Freedom of the World (EFW) index measures government size, legal systems and property rights, sound money, international trade, and regulation (Lawson et al., 2025). I summarize selected effects on firms through:

- $s \in (0, 1]$ is the **institutionally usable share of fixed property**: title disputes or restrictions can make premises unavailable for production.
- $c \geq 0$ is **recurring institutional overhead**, paid while open independently of output, such as permit renewals or fixed reporting costs.

Several EFW areas may affect either channel. These firm conditions are not index scores and do not cover every EFW component.

Experiment. The same firm starts open. Compare deterioration, no change, and improvement: $s_0 - d$, s_0, and $s_0 + d$ at date 0, with $0 < d < s_0$ and $s_0 + d \leq 1$. All paths return permanently to s_0 from date 1, as known in advance. Hold $c_t = c_0 \geq 0$ and all other parameters fixed.

Technology and choices. The pre-existing property stock is $\bar{k} > 0$; usable property is $s\bar{k}$. With productivity $A > 0$, elasticity $0 < \alpha < 1$, and variable input $x \geq 0$, assume Cobb–Douglas production. Choosing units so that $\bar{k} = 1$ gives

$$q = A(s\bar{k})^{1-\alpha}x^{\alpha} \quad \xrightarrow{\bar{k}=1} \quad q = As^{1-\alpha}x^{\alpha}. \tag{1}$$

For example, $s = 0.5$ makes half the fixed stock usable. Output price is one; input costs $w > 0$ per unit. Operating overhead is $F + c_t$: ordinary fixed costs $F > 0$ plus institutional expenses $c_t \geq 0$.

Activity is $a_t \in \{0, 1\}$: open is one, closed is zero; initially $a_{-1} = 1$. Closure is free and preserves property but dismantles the organization. Opening after closure costs $\kappa \geq 0$. Financing is available, and no temporary suspension option preserves the organization. The discount factor $0 < \beta < 1$ makes next period's unit of profit worth β today. The firm solves

$$\max_{\{a_t, x_t\}_{t\geq 0}} \sum_{t=0}^{\infty} \beta^t \left[As_t^{1-\alpha}x_t^{\alpha} - wx_t - (F + c_t)a_t - \kappa a_t(1 - a_{t-1})\right], \tag{2}$$

subject to $x_t \geq 0$ and $x_t = 0$ if $a_t = 0$. Each period contributes revenue minus input spending, operating overhead, and any restart payment. The last term charges κ only when the firm moves from closed to open; earlier setup costs have already been paid and cannot be recovered.

3 Expected first-order condition and output responses

Step 1: choose input when open. Today's input does not affect future production opportunities or restart costs. Operating profit can therefore be maximized separately in each period:

$$\pi(s, c) = \max_{x\geq 0}\{As^{1-\alpha}x^{\alpha} - wx - F - c\}. \tag{3}$$

Differentiating with respect to x gives the first-order condition (FOC):

$$\alpha As^{1-\alpha}x^{\alpha-1} - w = 0 \quad \Longrightarrow \quad \boxed{\alpha As^{1-\alpha}x^{\alpha-1} = w.} \tag{4}$$

Marginal revenue product equals input price. The second derivative is $\alpha(\alpha-1)As^{1-\alpha}x^{\alpha-2} < 0$; the slope falls from $+\infty$ to $-w$, giving a unique global optimum. Rearranging and substituting give

$$x^{1-\alpha} = \frac{\alpha As^{1-\alpha}}{w} \quad \Longrightarrow \quad x^*(s) = s\left(\frac{\alpha A}{w}\right)^{1/(1-\alpha)}, \tag{5}$$

$$q(s) = As^{1-\alpha}[x^*(s)]^{\alpha} = sA\left(\frac{\alpha A}{w}\right)^{\alpha/(1-\alpha)}. \tag{6}$$

Multiplying the FOC by x^* yields $wx^* = \alpha q(s)$; therefore

$$\pi(s, c) = q(s) - wx^*(s) - F - c = (1 - \alpha)q(s) - F - c. \tag{7}$$

Lower s reduces optimal input, operating output, and profit; higher c lowers profit without changing input conditional on operating. Actual output is $y_t = a_t q(s_t)$. **The FOC is unchanged across the baseline and extension:** the restart cost does not depend on input use. The changed prediction concerns activity and recovery, derived next.

Step 2: value operation and restarting. Define normal output $q_0 = q(s_0)$, normal profit $\pi_0 = \pi(s_0, c_0) > 0$, and shock profits $\pi_\pm = \pi(s_0 \pm d, c_0)$. Equation (6) gives

$$q(s_0 \pm d) = q_0 \left(1 \pm \frac{d}{s_0}\right), \qquad \pi_+ - \pi_0 = \pi_0 - \pi_-. \tag{8}$$

If the firm stays open, output and profit changes are equal and opposite. Asymmetry therefore comes from closure. After restoration, an active firm earns $\pi_0 > 0$ each period and has value

$$V = \pi_0 + \beta\pi_0 + \beta^2\pi_0 + \cdots = \frac{\pi_0}{1-\beta}. \tag{9}$$

A closed firm receives $V - \kappa$ from restarting or zero from staying out. Waiting $n \geq 0$ periods gives $\beta^n(V - \kappa)$: delay cannot improve on immediate profitable restart or make a loss profitable. During the deterioration at date 0, the firm compares

$$\underbrace{\pi_- + \beta V}_{\text{continue}} \qquad \text{and} \qquad \underbrace{\beta \max\{V - \kappa, 0\}}_{\text{close, retaining the restart option}}. \tag{10}$$

A current loss alone need not cause closure: future profits can justify staying.

Step 3: a conditional conjecture. Keep all assumptions in Section 2, including available financing, no suspension that preserves the organization, and the known institutional paths. Assume

$$\boxed{\pi_0 > 0, \qquad \pi_- + \beta V < 0, \qquad \kappa > \pi_+ + \beta V.} \tag{11}$$

Then the firm closes at date 0 and never restarts after deterioration; it stays open under no change or improvement. An otherwise identical inactive owner facing the same setup cost stays out in every path.

Argument. Since $\pi_+ > \pi_0$ and $\pi_0 + \beta V = V$, the last inequality implies $\kappa > V$, so closing is worth zero. Adverse continuation is negative; normal and favorable continuation are positive. Favorable entry yields $\pi_+ + \beta V - \kappa < 0$; less profitable or delayed entry cannot do better. The stronger cost bound excludes favorable entry; $\kappa > V$ alone suffices for the incumbent's no-restart decision.

Let y_t^-, y_t^0, and y_t^+ denote output under deterioration, no change, and improvement. Define the loss $L_t = y_t^0 - y_t^-$ and gain $G_t = y_t^+ - y_t^0$. Then

$$L_0 = q_0 > \frac{d}{s_0} q_0 = G_0, \qquad L_t = q_0 > 0 = G_t \quad (t \geq 1). \tag{12}$$

The first inequality uses $d < s_0$. Closure removes all normal output; improvement adds only a fraction. With **free restarting**, the same adverse firm closes because $\pi_- < 0$ but reopens at date 1 because $V > 0$. With costly restarting it remains closed: **full recovery becomes a persistent output shortfall**. Impact asymmetry can occur in both cases; restarting costs change persistence.

A closed firm's log output is undefined. To compare positive totals, suppose $Y_t^j = B + N y_t^j$, where $B > 0$ is unaffected output and $N > 0$ is a fixed number of identical firms. Hold B, N, and population fixed across paths. The arithmetic–geometric mean inequality gives

$$L_t \geq G_t \Rightarrow 2Y_t^0 \geq Y_t^- + Y_t^+ \geq 2\sqrt{Y_t^- Y_t^+} \Rightarrow \log Y_t^0 - \log Y_t^- \geq \log Y_t^+ - \log Y_t^0.$$

This establishes a conditional positive-output comparison, not an estimated GDP effect.

4 Related literature and proposed contribution

Dixit (1989) explains why a firm's decision depends on whether it already operates. A firm may tolerate losses to avoid rebuilding later; once it closes, recovery may not justify restarting. His discussion of organizational decay and temporary reversals (pp. 621, 623, 629) supplies the mechanism used here. Persistence through sunk costs is an established result.

Dixit (1992, pp. 119–120) distinguishes temporary suspension, which can preserve the business, from abandonment, which may require rebuilding. This identifies my model's main vulnerability: if waiting preserves the organization cheaply, institutional recovery may also restore production. The review includes this follow-up and its mathematical appendix.[1]

Dawson (1998) connects institutions, investment, and growth. At the firm level, Roberts and Tybout (1997) report evidence of sunk entry costs in Colombian export participation. This supports the relevance of costly entry, without establishing the EFW–GDP mechanism studied here.

I apply the established restarting mechanism to known institutional changes. The additional question is whether equal improvements and deteriorations produce unequal output responses. Costly restarting alone does not guarantee this direction. My contribution to the thesis is to state the conditions for the comparison, derive it, and test how suspension or new entry changes it. The proposal develops this application without claiming a new general theory.

5 Plan and risks

Prove by hand. I will show that the input choice is optimal, compare all feasible sequences of opening and closing, and derive (12) and the log comparison for positive total output. The argument above is a proof outline; a complete handwritten appendix remains future work.

Simulate. The current examples cover small shocks, free restarting, and costly restarting. I will vary normal margins, shock size, duration, and restart costs, and map continued operation, reopening, and permanent closure. I will then examine overhead shocks, suspension, entry, and heterogeneity. These are illustrative calculations, not empirical calibration.

Formalize in Lean. Lean will check that each conclusion follows from its stated assumptions. I will begin with $(1-\beta)V = \pi_0$, the opening and closing comparisons, and why delayed restarting cannot improve the firm's value. I will then formalize optimal input and the output inequalities, using AppliedModelingLib. This formal verification remains future work.

Most likely failure. A temporary setback may induce cheap suspension without dismantling the organization. That option can remove the lasting loss; it is the first robustness test. Entry and reallocation further challenge aggregate persistence. Passing numerical checks will not replace a proof.

References

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[1] The earlier working paper and separate 1989 appendices were unavailable. The review of Roberts and Tybout uses their published abstract.